

Introduction to Machine Learning

Project Title: Snake Species Classification: Venomous vs Non-Venomous using EfficientNet and Test-Time Augmentation (TTA).

By: -

Aarya Shinde (1132220237)

Harshwardhan Patil (1132220370)

1. Introduction:

India is home to many snake species, both venomous and non-venomous. The ability to correctly classify these snakes is critical in medical, environmental, and wildlife rescue fields. This project aims to develop a **Machine Learning model using Deep Learning (CNN)** to classify snake images into **venomous** or **non-venomous** categories using **EfficientNet-B0** architecture. It incorporates **Test-Time Augmentation (TTA)** for improving generalization and reliability.

2. Problem Statement:

Due to the visual similarity between some venomous and non-venomous snakes, traditional classification methods struggle with accuracy. Our objective is to:

- Automate snake species classification using images.
- Ensure fast and reliable predictions.
- Handle class imbalance using augmentation and model tuning.

3. Objectives:

- Build an image classification model to distinguish between venomous and non-venomous snakes.
- Utilize GPU acceleration (CUDA) for faster training.
- Improve prediction reliability with Test-Time Augmentation.
- Visualize predictions with confidence scores and image output.

4. Dataset Description:

The dataset is a structured folder of images categorized into two main classes:

- **Train Set:** Used for model training, includes both venomous and non-venomous classes.
- **Test Set:** Used for model evaluation.

5. Technologies Used:

- **Language:** Python
- **Libraries:** PyTorch, torchvision, scikit-learn, matplotlib, PIL
- **Model:** EfficientNet-B0 (Pretrained)
- **Augmentation:** Horizontal Flip, Rotation, Normalization
- **Hardware Acceleration:** CUDA 12.1.1 with cuDNN 8.9.0.131

6. Model Architecture:

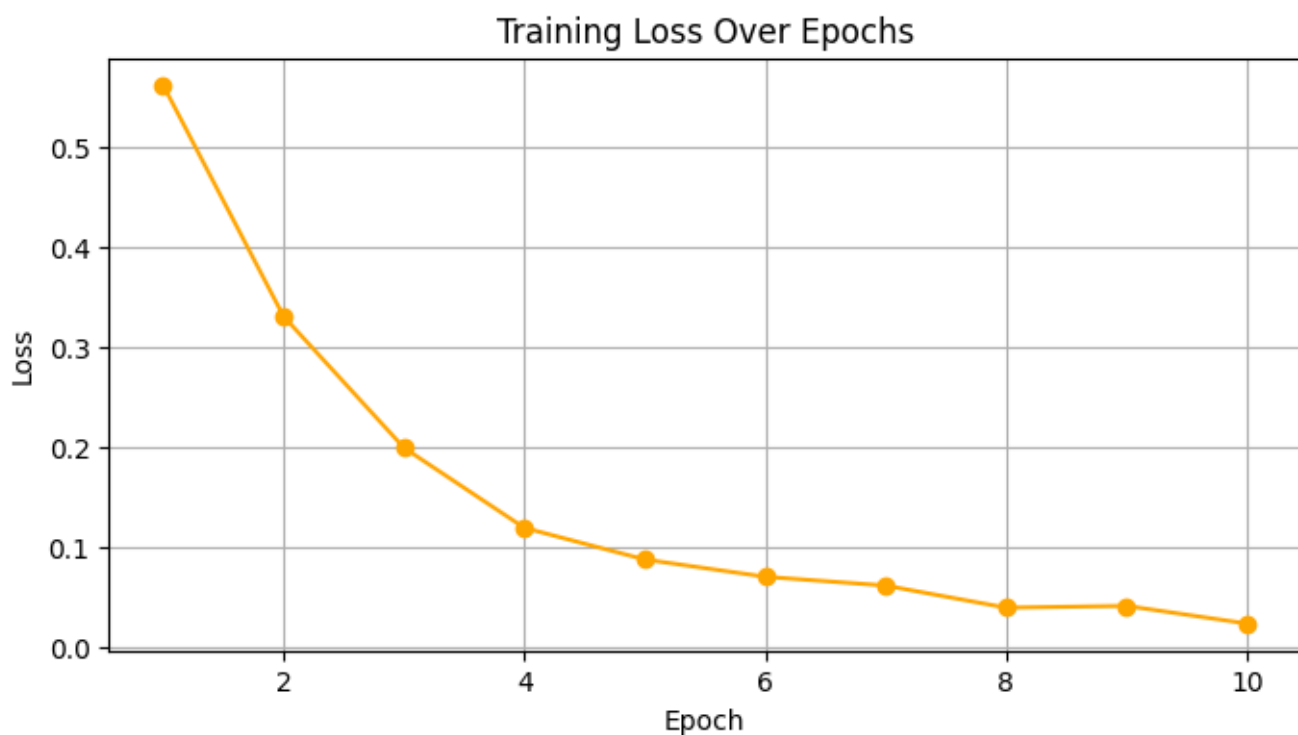
We used **EfficientNet-B0**, a convolutional neural network known for high performance with fewer parameters. It's fine-tuned on our custom dataset by replacing the final classification layer with a 2-class output layer.

7. Test-Time Augmentation (TTA):

Test-Time Augmentation is a technique used during inference time to improve prediction reliability. Instead of predicting directly on a single image, we generate several augmented versions of that image (e.g., horizontally flipped, rotated), make predictions for each version, and then average the results.

Benefits:

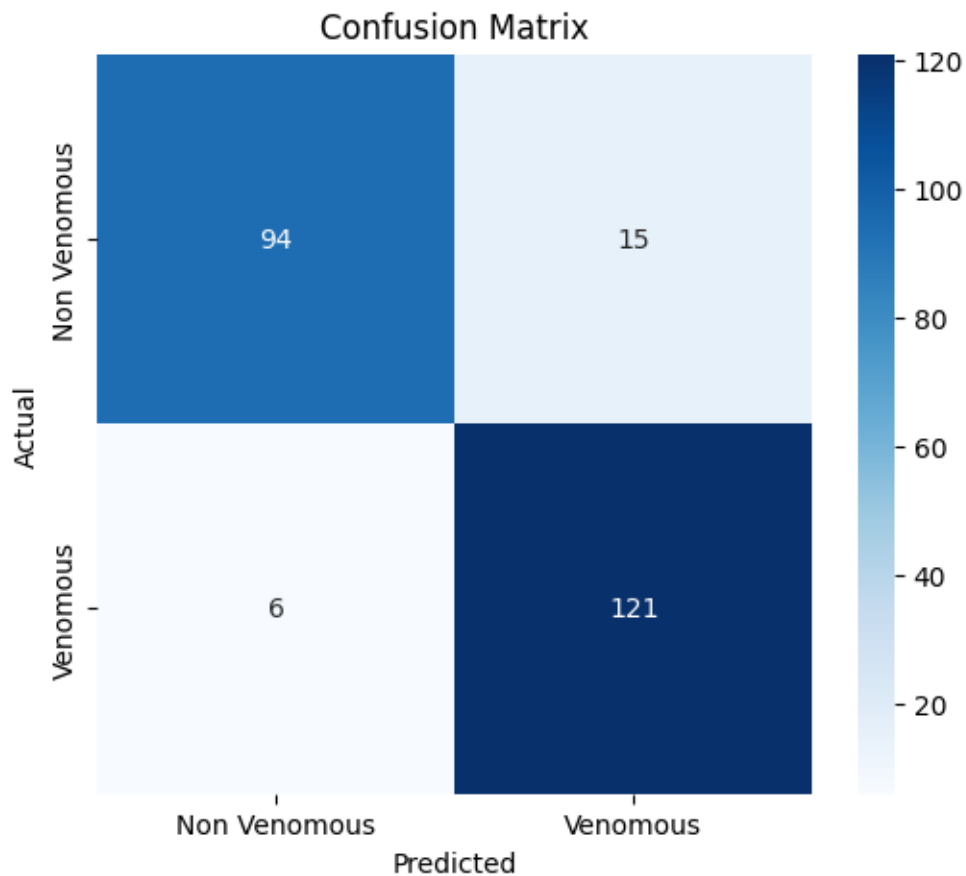
- Helps mitigate overfitting.
- More confident and stable predictions.
- Can simulate ensemble-like robustness without training multiple models.



9. Evaluation Metrics:

	Precision	Recall	Support	F1-Score
Non Venomous	0.94	0.86	0.90	109
Venomous	0.89	0.95	0.92	127

Confusion Matrix



Prediction with Random Data (outside of Test/Train Dataset) :-

Choose Image



Prediction: Venomous (1.00)

⚠️ DANGER! Venomous Snake Detected! ⚠️
Seek immediate medical attention!

Train Model

Additional Features

1. Graphical User Interface (GUI):

The system includes a user-friendly and interactive GUI developed using libraries such as **Tkinter** or **PyQt**. Through this interface, users can:

- Upload snake images directly from their device.
- View the predicted class: **Venomous** or **Non-Venomous**.
- See confidence scores for each prediction.
- Monitor model performance metrics like accuracy, precision, and recall.
- Access a history of previous uploads and predictions.

2. Voice Alert System:

The system also features a **voice-based alert mechanism** implemented using Python's `pyttsx3` or similar libraries. It:

- Provides **instant voice feedback** on the classification outcome.
- Clearly announces whether the identified snake is **venomous** or **non-venomous**.
- Delivers a warning alert for venomous snakes to ensure quick response and caution.

These features enhance the application's practicality and usability, especially for forest officers, hikers, wildlife researchers, and emergency responders dealing with snakebite risks.

11. Conclusion:

The model achieved a high test accuracy of **91.10%**, effectively classifying snake images. It utilizes GPU computing, advanced CNN architecture, and TTA for enhanced performance. This project can be expanded to support more classes (species-wise identification) or real-time applications for rescue or educational tools.

12. Future Scope:

- Expand to multi-class classification.
- Integrate into mobile applications.
- Real-time video detection and alert system.

